

Kriging Assisted Surrogate Evolutionary Computation to Solve Optimal Power Flow Problems

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Abstract—This paper proposes a Kriging assisted strategy to expedite evolutionary computation for solving Optimal Power Flow (OPF) problems. First, two algorithms were developed—a Kriging Assisted Genetic Algorithm (KAGA) and a Kriging Assisted Particle Swarm Optimization (KAPSO) - and tested using unconstrained benchmark functions; it was found that both algorithms provided reliable and robust solutions. Accordingly, KAGA and KAPSO were selected and tested on the IEEE 30 and 118 bus systems for minimizing generation costs and active power losses. It is shown that the proposed KAPSO outperforms other algorithms, especially in terms of the computation time. In reference to the solution quality yielded by KAGA and KAPSO, the proposed Kriging assisted strategy offers a promising method to improve the performance of evolutionary based computation when solving OPF problems.

Index Terms—Optimal power flow, genetic algorithm, particle swarm optimization, Kriging, surrogate modelling, evolutionary computation.

I. INTRODUCTION

SINCE the concept of Optimal Power Flow (OPF) was first introduced as a Non-Linear Programming (NLP) problem by Carpentier [1] for solving Economic Dispatch (ED) problems, it has been used as an essential tool in planning, operating, and dealing with energy management aspects of power systems. The complexity of the problem depends critically on the selection of objective functions and design variables due to the combinatorial nature of the components and their connections. Mathematically, the formulation leads to a large-scale, non-convex, non-linear, highly constrained optimization problem where the continuous (e.g., generator voltage magnitudes, generator active power outputs) and discrete (e.g., transformer taps, switch shunts) variables are optimized within specified constraints to optimize the desired objectives, such as generation cost, active power loss, improved transient stability, etc.

Over the years Linear Programming (LP) [2], Quadratic Programming (QP) [3], Mix-Integer Linear Programming (MILP) [4] and Mix-Integer Non-Linear Programming (MINLP) [5]

have been proposed to efficiently define the OPF problems, while the selection of the programming formulation usually implicates the solution method [6]. For example, LP requires linearizing the objective function and constraints, whereas QP is a variant of NLP in which the objective function is expressed as a quadratic form and the constraints are linearized. The deterministic methods—like the Simplex Method [7] and Sequential Linear Programming Method [2], Sequential Quadratic Programming Method [8], Newton Based Method [9] and Interior Point Method (IPM) [10], have been applied for many years to solve the practical OPF problems in the form of LP, QP and NLP, respectively. Because the constraints are efficiently handled (e.g., IPM converts a constrained OPF problem into an unconstrained one by introducing a Lagrangian objective function) and they produce reasonable optimal solutions (e.g., Karush-Kuhn-Tucker conditions) at excellent computational speed. These methods, however, rely strongly on the convexity of the problem (while OPF problems are naturally non-convex) and the choice of an initial starting point in the search for an optimal solution, thus do not guarantee to converge to a global optimum. Moreover, although suitable MILP and especially MINLP forms have been proposed to accurately capture the behavior of the OPF problems, deterministic methods may not be appropriate for the solution of complex optimization tasks. This is due to the fact that deterministic methods generally consider the OPF problems as being continuous, cannot easily handle discrete variables, and the derivative of the objective function may not always be available. To use a deterministic method, a compromise may have to be made in terms of reliability, accuracy and computation efficiency [11]. Recent years, evolutionary based algorithms (i.e., non-deterministic methods) have attracted attention in academic research, because they can easily deal with the drawbacks of the deterministic methods. Theoretically, the global optimum can always be found after many iterations by using the non-deterministic methods—such as Genetic Algorithm (GA) [12], Particle Swarm Optimization (PSO) [13] and Simulated Annealing (SA) [14]. However, the computational cost dramatically increases due to a large number of calls for objective function evaluation (e.g., full AC power flow analysis) and this limits the application of evolutionary computation in real-time OPF situations.

All reported OPF methodologies, including deterministic and non-deterministic, have advantages in some aspects and weaknesses in others [11]. Surrogate modelling techniques were

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introduced some time ago in optimization community but have not been applied to solve OPF problems. The focus on surrogate assisted evolutionary optimization for accelerating the overall computational speed by reducing the total number of calls to objective function evaluation started in 2002 during the Genetic and Evolutionary Computation Conference (GECCO) [15]. To summarize, the strategies for combining surrogates with evolutionary computation were classified into three categories: individual-based, population-based and generation-based [16]. Thus the fitness of some individuals, populations or generations is evaluated using a real objective function, while others are estimated by a surrogate model. By properly managing a surrogate model the number of function evaluations can be reduced and thus the CPU time improved. A variety of surrogate modelling methods exist, including Polynomial Regression, Radial Basis Function, Neural Networks and Kriging (also known as Gaussian Processes). Kriging proved to be the most successful surrogate method in electromagnetic design as argued and reported by the Southampton group [16], [17]. However, the application of Kriging has so far been limited to relatively small-scale optimization tasks, whereas OPF problems tend to have a large number of variables. The issue arises due to the need to invert a large correlation matrix with associated $O(n^3)$ computational cost and $O(n^3)$ storage requirements [19], where n is the sample size. Furthermore, the low fidelity of a surrogate model may cause the evolutionary optimization to fall into a local optimum trap.

This paper exploits the Kriging method to reduce the computational burden associated with evolutionary solutions of OPF problems. An individual-focused strategy—which efficiently addresses the fidelity, computational cost and storage requirements of using Kriging in combination with evolutionary computation—is proposed. A Euclidean distance based method is developed to balance exploitation and exploration when selecting a particle (individual) for updating the population and the Kriging sample set. To illustrate the strategy, a Kriging Assisted Genetic Algorithm (KAGA) and a Kriging Assisted Particle Swarm Optimization (KAPSO) are developed and tested on four benchmark functions with results compared with the GA and the PSO. KAGA and KAPSO are then chosen due to their reliable and robust performance and applied to the IEEE 30 and 118 bus systems for minimizing the generation cost and active power losses. The ensuing solutions are compared with published results.

II. OPTIMAL POWER FLOW FORMULATION

The classic NLP formulation can be written as

$$\min. F(\mathbf{x}, \mathbf{u}) \quad (1)$$

$$\text{s.t. } \mathbf{g}(\mathbf{x}, \mathbf{u}) = 0 \quad (2)$$

$$\mathbf{h}(\mathbf{x}, \mathbf{u}) \leq 0 \quad (3)$$

where $F(\mathbf{x}, \mathbf{u})$ is the objective function to be minimized by adjusting the control variables $\mathbf{u}$ while satisfying the equality constraints $\mathbf{g}(\mathbf{x}, \mathbf{u})$ and the inequality constraints $\mathbf{h}(\mathbf{x}, \mathbf{u})$.

The vectors of control variables $\mathbf{u}$ and state variables $\mathbf{x}$ are

$$\mathbf{x} = [P_{G1}, V_{d1} \cdots V_{dn}, \delta_2 \cdots \delta_n, Q_{G1} \cdots Q_{Gn}] \quad (4)$$

$$\mathbf{u} = [P_{G2} \cdots P_{Gn}, V_{G1} \cdots V_{Gn}, T_1 \cdots T_l, Q_{C1} \cdots Q_{Cn}] \quad (5)$$

where the state variables contain the active power output of the slack bus generator (P_{G1}), voltage magnitudes of load buses (V_{dn}), voltage angles (δ_n) of all buses except the slack bus, and the reactive power output of each generator; the control variables include the active power outputs P_{Gn} of all generators except the slack bus generator, the voltage magnitude (V_{Gn}) of all generators, the transformer taps, and the switchable shunt capacitors, where l and n are the number of transmission lines and buses, respectively.

A. Objective Functions

The proposed algorithm could be applied to a variety of OPF problems; two general objective functions are studied in this paper:

1) *Generation Cost*: The generation cost is one of the main concerns of a system operator for achieving optimal economic dispatch during the regular system operation, and may be expressed as

$$F_{\text{cost}} = \sum_{i=1}^n a_i P_{Gi}^2 + b_i P_{Gi} + c_i \quad (6)$$

where a_i , b_i , and c_i and P_{Gi} are the cost coefficients and active power output of the i th generator, respectively, for calculating the total generation cost. Here the standard quadratic cost function is used, while the other cost functions—like piecewise quadratic cost function and quadratic cost curve with valve-point effect—may be found in [20].

2) *Active Power Losses*: Large amounts of electricity can be saved by minimizing active power losses and it is important to balance the system under a heavy loaded situation; this can be written as

$$F_{\text{losses}} = \sum_{\substack{i=1 \\ k \neq i}}^n [V_i^2 - V_k^2 - 2V_i V_k \cos(\delta_i - \delta_k)] G_{ik} \quad (7)$$

where G_{ik} is the line conductance between two buses, V_i and the V_k are the voltage magnitudes at the sending bus i and the receiving bus k , while δ_i and δ_k are the voltage angles at the i th and k th bus, respectively.

B. Constraints

1) *Equality Constraints*: For the system operator, the top priority is to ensure that the power generation capacity meets the demand (including the losses). Thus the active and reactive power flow balance equations are usually considered as equality

constraints for the OPF problems

$$P_i = P_{Gi} - P_{Di} = \sum_{k=1}^n V_i V_k Y_{ik} \cos(\theta_{ik} - \delta_i + \delta_k) \quad (8)$$

$$Q_i = Q_{Gi} - Q_{Di} = \sum_{k=1}^n V_i V_k Y_{ik} \sin(\theta_{ik} - \delta_i + \delta_k) \quad (9)$$

where P_i and Q_i are the injected power at bus i , P_{Gi} and Q_{Gi} are the generated active and reactive power at bus i , P_{Di} and Q_{Di} are the demand active and reactive power at bus i , and θ_{ik} is the line admittance angle between the two buses. Ideally, the injected power at the bus should just meet the demand power, resulting in balanced power flow. In this study, the balance equations are considered as a power flow problem and calculated using the Newton-Raphson method.

2) *Inequality Constraints*: In order to make the system secure, the state variables and control variables should fall within their limits. To achieve this the following inequality constraints are prescribed

$$P_{Gi}^{\min} \leq P_{Gi} \leq P_{Gi}^{\max} \quad (10)$$

$$Q_{Gi}^{\min} \leq Q_{Gi} \leq Q_{Gi}^{\max} \quad (11)$$

$$V_i^{\min} \leq V_i \leq V_i^{\max} \quad (12)$$

$$T_l^{\min} \leq T_l \leq T_l^{\max} \quad (13)$$

$$Q_{Ci}^{\min} \leq Q_{Ci} \leq Q_{Ci}^{\max} \quad (14)$$

where the terms in (10), (11) and (12) for expressing the slack bus active power output limits, reactive power output limits and bus voltage magnitude limits, respectively, indicate the degree of reliability and security after the control variables have been adjusted (i.e., state variable limits). The other equations specify the acceptable ranges of control variables. Moreover, these variables and limits are assigned practical values when applied to the tested optimization cases. Consequently, only the state variable constraints are actually applied when using evolutionary based algorithms, because the control variables are always encoded within the feasible ranges during the optimization process. Finally, although only continuous variables are considered here, the discrete control variables can also be readily handled by the proposed method as in the non-deterministic methods [5].

C. Penalty Function

To impose the above state variable constraints, a penalty function is added to the objective function and expressed as

$$F = F + \lambda \quad (15)$$

$$\lambda = k \sum_i^n f(x_i) \quad (16)$$

$$f(x) = \begin{cases} x^{\min} - x, & \text{if } x < x^{\min} \\ x - x^{\max}, & \text{if } x > x^{\max} \\ 0, & \text{if } x^{\max} \geq x \geq x^{\min} \end{cases} \quad (17)$$

where k is a self-defined value representing the penalty weighting factor, while (17) is a fuzzy decision-making function to calculate the penalty function. Note that the constraints of control variables can be easily handled by any evolutionary based algorithm, while the penalty function is mainly used here for the state variables described by (4).

III. KRIGING ASSISTED EVOLUTIONARY COMPUTATION

Amongst various non-deterministic methods, GA and PSO are best known with proven computational efficiency. The methods will be mentioned only briefly as a more detailed description may be found for example in [21]. A combination of Kriging with GA and PSO is the proposed novel approach here, but it should be emphasized that this new strategy may also be applied to other evolutionary algorithms.

A. Kriging

Kriging, also known as the stochastic Gaussian Process (GP), uses observed data (samples) to create a response surface to estimate the values of the objective function at unobserved locations. This process may be expressed as

$$y^*(x^*) = \mu + \epsilon(x^*) \quad (18)$$

where x^* is the new point to be estimated, $y^*(x^*)$ is the estimated value at point x^* , μ is the unknown mean value of the GP, and $\epsilon(x^*)$ is assumed to be the normally distributed noise (or errors) with the mean value of zero and variance σ^2 . Intuitively, this equation can be interpreted as an estimated value at x^* which is uncertain and likely to vary within some range ($\mu \pm n\sigma$). Generally, the exact errors of the GP are unknown, but Kriging assumes the errors between any two points to be correlated as

$$\text{corr}[\epsilon(x_i), \epsilon(x_k)] = \prod_{d=1}^N \exp(-\theta_d |x_{id} - x_{kd}|^{p_d}) \quad (19)$$

where d is the d th dimension of variables x , N is the size of the dimensions, the hyperparameter θ_d adjusting the Kriging model describes how fast the correlation drops away from one point to another in the d th dimension, and the values of p_d in general are taken to be equal to 2, which helps to smooth a line between any two points and thus the whole response surface.

The deviations of the stochastic GP are quantified by the unknown parameters μ , σ^2 and θ_d , and the likelihood function of the observed data is given by

$$\mathcal{L}(y|\mu, \sigma^2) = \left[-\frac{n}{2} \log(\sigma^2) - \frac{1}{2} \log(|R|) \right] - \left(\left[\frac{(y - 1\mu)' R^{-1} (y - 1\mu)}{2\sigma^2} \right] \right) \quad (20)$$

where y is the vector of observed data, R is a $n \times n$ (n is the number of samples) correlation matrix which—in terms of the (i, k) elements (for $i, k = 1, \dots, n$)—provides the calculated pairwise correlations between pairs of observed data points from (19). To achieve consistency of the Kriging model with the observed data, the optimal values of unknown parameters μ

and σ^2 are found by maximizing the likelihood function to the observed data as

$$\hat{\mu} = \frac{1' \mathbf{R}^{-1} \mathbf{y}}{1' \mathbf{R}^{-1} \mathbf{1}} \quad (21)$$

$$\hat{\sigma}^2 = \frac{(\mathbf{y} - 1\hat{\mu})' \mathbf{R}^{-1} (\mathbf{y} - 1\hat{\mu})}{n} \quad (22)$$

which have been derived through partial differentiation of the likelihood function with respect to each parameter subject to the necessary condition (i.e., zero). By substituting (21) and (22) into (20) and ignoring the last term (which does not depend on θ_d), the likelihood function with respect to θ_d may be derived. This could be considered as a maximization problem so that the optimal $\hat{\theta}_d$ may be found using any optimization algorithm. Note, however, that the lower and upper bounds of $\hat{\theta}_d$ need to be carefully defined, especially as a large $\hat{\theta}_d$ may result in a highly sparse correlation matrix that makes the value of any estimated point away from the observed points to be approximated to the mean value of the observed values. Here, the lower and upper bounds were set to 0.001 and 1, respectively.

To estimate the values at a new point $\mathbf{x}^*$, the predicted values $(\mathbf{x}^*, \mathbf{y}^*)$ are added to the observed data to form an augmented set, defined as $(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = [(\mathbf{x}, \mathbf{x}^*)', (\mathbf{y}, \mathbf{y}^*)']$. Thus the correlation matrix for the augmented data is given as

$$\hat{\mathbf{R}} = \begin{bmatrix} \mathbf{R} & \mathbf{r} \\ \mathbf{r}' & 1 \end{bmatrix} \quad (23)$$

where $\mathbf{r}$ is a $n \times m$ (m is the number of estimated points) correlation matrix with (i, j) elements ($i = 1, \dots, n$, and $j = 1, \dots, m$), containing calculated pairwise distances between estimated points and observed points, similar to (19). Referring to (20), as only the last term depends on augmented data, the augmented likelihood function becomes

$$\begin{aligned} \tilde{\mathcal{L}}(\mathbf{y}|\hat{\mu}, \hat{\sigma}^2, \mathbf{y}^*) &= \left(\frac{-1}{2\hat{\sigma}^2 (1 - \mathbf{r}' \mathbf{R}^{-1} \mathbf{r})} \right) (\mathbf{y}^* - \hat{\mu})^2 \\ &+ \left(\frac{\mathbf{r}' \mathbf{R}^{-1} (\mathbf{y} - 1\hat{\mu})}{\hat{\sigma}^2 (1 - \mathbf{r}' \mathbf{R}^{-1} \mathbf{r})} \right) (\mathbf{y}^* - \hat{\mu}) \end{aligned} \quad (24)$$

and—as a function of $\mathbf{y}^*$ —measures the consistency of estimated values with observed data. A more comprehensive derivation of (24) may be found in [22]. By taking the partial derivative of (24) with respect to $\mathbf{y}^*$ and equating to zero to maximize the likelihood, the standard Kriging estimator is

$$\mathbf{y}^*(\mathbf{x}^*) = \hat{\mu} + \mathbf{r}' \mathbf{R}^{-1} (\mathbf{y} - 1\hat{\mu}) \quad (25)$$

where the estimated $\mathbf{y}^*$ is the most confident prediction given by the Kriging estimator. It can be seen that if θ_d is too large, the elements in $\mathbf{r}$ and $\mathbf{R}$ will be close to zero, and thus the estimated $\mathbf{y}^*$ approximated to $\hat{\mu}$.

By using the optimized variance ($\hat{\sigma}^2$) of the Kriging model [23], the Mean Squared Error (MSE) of Kriging is given by

$$\hat{\mathcal{S}}^2(\mathbf{x}^*) = \hat{\sigma}^2 \left[1 - \mathbf{r}' \mathbf{R}^{-1} \mathbf{r} + \frac{(1 - 1' \mathbf{R}^{-1} \mathbf{r})^2}{1' \mathbf{R}^{-1} \mathbf{1}} \right]. \quad (26)$$

Note that this is not a true error between the estimated and real values, but an implied potential error (or uncertainty) of the prediction. From the statistics point of view, the MSE is

inversely related to the curvature of the augmented likelihood function. Thus the larger the MSE, the lower the confidence of the prediction and vice versa.

B. Proposed Individual-Based Strategy

1) *Initialization*: This defines the population size, the maximum number of iterations and the essential parameters for GA or PSO. Then a real-valued population is randomly generated consisting of a specified number of individuals whose fitness is evaluated; this is later used to build a Kriging model.

2) *Updating the Individuals*: For GA, the arithmetic crossover [24] and Gaussian mutation [24], [25] operators are employed to update the individuals; for PSO, the type 1'' constriction coefficients [27] are used to calculate the velocity of each particle for updating the swarm.

3) *Evaluate the Fitness*: The Kriging model is constructed using the given samples (i.e., initial population), for estimating the fitness and MSE of all updated individuals. A Bayesian based technique called Expected Improvement (EI) [28] is applied to guide the selection of which individuals should be evaluated using the real objective function. The calculation of EI is based on the estimated MSE of individuals as

$$\begin{aligned} E[\mathbf{I}(\mathbf{x}^*)] &= \mathbf{w}(\mathbf{x}^*) [y_{best} - \mathbf{y}^*(\mathbf{x}^*)] \Phi \left(\frac{y_{best} - \mathbf{y}^*(\mathbf{x}^*)}{\hat{\mathcal{S}}(\mathbf{x}^*)} \right) \\ &+ (1 - \mathbf{w}(\mathbf{x}^*)) \hat{\mathcal{S}}(\mathbf{x}^*) \phi \left(\frac{y_{best} - \mathbf{y}^*(\mathbf{x}^*)}{\hat{\mathcal{S}}(\mathbf{x}^*)} \right) \end{aligned} \quad (27)$$

where Φ and ϕ denote the standard normal cumulative distribution function and probability distribution function, respectively, while y_{best} is the current best objective value (minimum or maximum for minimization or maximization problems, respectively) and $\hat{\mathcal{S}}$ is the square root of MSE (which itself reflects the uncertainty of the estimated value). If the estimation at $\mathbf{x}^*$ is assumed to be normally distributed with the mean $\mathbf{y}^*$ and standard deviation $\hat{\mathcal{S}}$, this equation can be interpreted as follows: the first term is large when Kriging estimation is likely smaller than the current minimum, whilst the second term is large when the Kriging estimation is with high uncertainty (large MSE). The calculated EI can also be considered as a compromise between exploitation (first term) and exploration (second term), where $\mathbf{w}(\mathbf{x})$ is the weighting factor to balance the exploitation and exploration. In this paper, a simple and effective method based on Euclidean distance, as inspired by [29], is proposed to be

$$\mathbf{D}_i = \|\mathbf{x}_i^* - \mathbf{x}_{best}\| \quad (28)$$

$$\mathbf{D}_i^{normalized} = \frac{\mathbf{D}_i - \min(\mathbf{D}_i)}{\max(\mathbf{D}_i) - \min(\mathbf{D}_i)} \in (0, 1) \quad (29)$$

$$\mathbf{w}(\mathbf{x}_i^*) = \frac{1}{1 + 9 \exp(-4.4 \mathbf{D}_i^{normalized})} \in (0.1, 0.9) \quad (30)$$

where $\mathbf{x}_{best}$ is the position of the current minimum, (28) is a Euclidean metric used to measure the distance between each individual and the current minimum, (29) normalizes the measured distances within the range from zero to one, and (30)

calculates the weighting factor for each individual. Substituting the calculated $w(x_i^*)$ into (27) puts more emphases on exploration if the individual is close to the current minimum, or on exploitation if the individual is far from the current best. Finally, the individual/particle which has the maximum value of EI (later called the ‘infilling point’) is selected and evaluated using real objective function.

4) *Updating the Population and Samples*: A possible strategy, as recommended in [16], is to have a certain number of individuals, based on the estimated fitness, that are selected and re-evaluated using the real objective function for updating the population. However, this may push the optimization process towards a local optimum and result in unwanted additional FEs, should the constructed surrogate model have low fidelity or be over-fitted. We propose that the infilling point should replace the individual (for KAGA) or personal best (for KAPSO) with the worst fitness. Even if the fitness of an infilling point is not better than the current worst fitness of an individual or the personal best, a point selected by the maximum EI should be used as a replacement because of the information it contains.

In general, the infilling points need to be iteratively added into the samples to build a global Kriging model; however, this might significantly increase the computational cost and require large amounts of physical memory, due to the need to invert a big correlation matrix as seen from (25) and (26). In our proposed strategy, the infilling point will replace the best sample point only if the infilling fitness is better than or equal to the best sample; otherwise, the infilling point will replace the sample with a worst fitness. This is claimed to be an effective way to alleviate the computing issues of using Kriging, and a local surrogate model around the optimal solution region could be eventually constructed.

5) *Elitist Learning Strategy*: In order to avoid a local optimum, a mutation strategy adopted from [29] is used to ‘jump out’ of the local optimum

$$x_{mutation} = x_{best}^d + (\bar{x}^d - \underline{x}^d) * normal(0, sigma) \quad (31)$$

where d is randomly selected so that every dimension has equal probability to be mutated, $\bar{x}^d$ and $\underline{x}^d$ denote the upper and lower bounds of the variable in the d th dimension, and $normal(0, sigma)$ is a normal random number with the mean zero and standard deviation ‘sigma’. The ‘sigma’ is generally set to decrease linearly along with the process of iterations from 1 to 0.1. However, it may be (although this tends to be problem dependent) that a constant value results in a better convergence. In this Elitist Learning Strategy (ELS), only one dimension is mutated at a time because the local optimum may actually belong to the global best. After evaluating the mutated local best position, the population—or the individuals, the personal best and the sample—are updated using the same procedure as in step 4. Moreover, in a similar way as when updating the population, the mutated individual (or particle) will be used even if it possesses a poor fitness than the current worst fitness in the population, because of its usefulness in global searching.

The proposed algorithm then goes back to step 2 and continues until the predefined maximum iteration number is reached or another termination criterion is met (e.g., the standard deviation of a sample smaller than a defined value).

C. Tests Using Benchmark Optimization Functions

Before applying to real OPF problems the two algorithms were tested using four multimodal benchmark functions (Ackley, Levy, Rastrigin and Schwefel functions, details of which may be found in [30]). All test functions used 50 variables and their accurate optimal solutions are all equal zero. The GA and PSO were implemented using the open source and well-developed Matlab codes from [31], whereas the KAGA and KAPSO were implemented by integrating GA and PSO with DACE (Kriging) [32]. The population size and the maximum number of iterations for these test algorithms were set to 20 and 50,000, respectively. In order to balance exploitation and exploration, the crossover probability of each selected parents and the mutation probability of each offspring for GA and KAGA were 0.7 and 0.3, respectively, while for KAPSO constriction coefficients were used. Moreover, mutation is performed by randomly selecting $ceil(0.01 * nVar)$ variables in the offspring and generating new values based on Gaussian distribution [24], [25], where $ceil$ is the ceiling operator and $nVar$ is the total number of variables. Finally, the ELS was employed in the standard PSO to avoid the premature convergence.

The results in terms of the mean Optimal Solution (OS), average number of Function Evaluations (FEs) required to find the OS within the specified tolerance of 0.01, and the total Success Rate (SR) of finding an acceptable optimal solution for 50 trials obtained from GA, PSO, KAGA and KAPSO are provided in Table I, where ‘INF’ means that the algorithm failed to converge to the optimum within the specified tolerance after reaching the maximum number of allowed iterations. Comparing unassisted GA and PSO with the versions utilizing Kriging reveals that using this type of surrogate modeling always improves the performance, in terms of Success Rate, finding a better optimum and using smaller number of necessary function evaluations, sometimes significantly. Note that the Kriging assisted approaches were actually consuming more computational time than the GA and PSO in these tests, because the test functions are computationally very cheap, while the Kriging assisted strategies were introduced mainly for solving problems which have expensive objective function evaluations (e.g., Finite Element Model). However, it has to be emphasized that the more complex the objective function evaluation is, the more efficient the Kriging assisted methods tend to be. Furthermore, the GA and PSO were visibly trapped in local minima for some test functions (e.g., Schwefel and Ackley have low success rates), whereas the assistance of Kriging enabled a global optimum to be easily found. Kriging may also be helpful in combination with other optimization algorithms; GA and PSO were chosen here because of their popularity and availability of relevant codes. Finally, comparing KAGA with KAPSO suggests that overall both proposed methods provide similar quality answers; thus it was decided to use KAGA and KAPSO for the OPF problems addressed in this paper.

IV. OPF RESULTS AND DISCUSSION

For the OPF analysis, the proposed KAPSO algorithm was implemented by integrating GA/PSO [31], DACE (Kriging)

TABLE I
MEAN SOLUTIONS, MEAN FUNCTION EVALUATIONS AND SUCCESS RATE OF THE 50 TRIALS PERFORMED BY DIFFERENT ALGORITHMS

Algorithms	Ackley Function			Levy Function			Rastrigin Function			Schwefel Function		
	OS	FES	SR	OS	FES	SR	OS	FES	SR	OS	FES	SR
GA	1.89E-2	8.54E+5	10	1.03E-5	1.19E+5	100	8.40E-3	6.35E+5	64	4.75E-2	7.20E+5	2
PSO	1.42E-3	4.02E+5	94	9.50E-6	1.33E+5	100	5.23E-3	6.56E+5	88	2.9028	INF	0
KAGA	3.97E-7	2.42E+4	100	3.71E-14	8.04E+3	100	1.35E-6	5.12E+4	100	6.36E-4	2.76E+4	100
KAPSO	2.03E-6	5.58E+4	100	1.03E-17	1.31E+4	100	4.52E-9	5.24E+4	100	6.36E-4	2.30E+4	100

[32], and MATPOWER [33] in MALAB 2017a on a PC (Intel Core i7-4790@3.6 GHz, 16 GB memory) for minimizing generation cost (Case 1) and active power losses (Case 2) of the IEEE 30 and 118 bus systems. Detailed information about the test systems may be found in MATPOWER casefiles. The upper and lower tap ratings are 1.1 and 0.9. Other GA, PSO KAGA and KAPSO parameter settings were the same as for the benchmark functions of Section III, except for the population size and the maximum number of iterations which were chosen in accordance with the size of the test problems. Moreover, the number of FEs (i.e., power flow calculations) was selected to be the additional index for comparing the computational performance, rather than only the elapsed time which might be affected by differences in hardware or compilers. Furthermore, the OPF problems were assumed to be continuous, with no discrete variables. In addition, the ELS was applied to both GA and PSO in this study to prevent being trapped in local optima. Finally, to avoid the ELS becoming too aggressive or too conservative, the ‘sigma’ in (31) was set to 0.1.

A. IEEE 30 Bus System

The system has 41 transmission lines, 283.4 MW active and 126.2MVar reactive power demand, six generators at buses 1, 2, 5, 8, 11 and 13 (with the generator at bus 1 taken as a swing generator), four transformers installed between buses 6–9, 6–10, 4–12 and 27–28, and nine shunt capacitors at buses 10, 12, 15, 17, 20, 21, 23, 24 and 29. The shunt compensators are adjusted between 0 and 5 MVar [28]. The acceptable voltage magnitudes of generator buses considered can vary between 1.1 and 0.95 pu, whereas the lower and upper bounds for load buses were set to 1.05 and 0.95, respectively. In order to ensure the state variables remain within the limits, the penalty factor k described in (16) was set to 1000 and 1 for load bus voltage magnitude and reactive power generation constraints respectively. The population size for KAGA and KAPSO needs to be carefully selected, as the larger the population size the better the accuracy of the Kriging model; a large population, however, will increase the computation time because big correlation matrices need inversion. For this small-scale system the population size between 20 and 50 was considered suitable and 50 was actually used. The maximum number of iterations for KAGA and KAPSO was set to 500 (as only two FEs are required at each iteration), a larger value than the one used in GA and PSO (i.e., 100), to ensure the global optimum was found.

The best optimal settings of 50 trials obtained by KAGA and KAPSO for Cases 1 are given in Table II, and the optimal

TABLE II
BEST OPTIMAL SOLUTIONS FROM KAGA AND KAPSO FOR MINIMIZING COST

Variables	KAGA	KAPSO	Variables	KAGA	KAPSO
P_{g2} (MW)	48.8514	48.7419	T_{6-10}	0.9119	0.9581
P_{g5} (MW)	21.3771	21.2718	T_{4-12}	0.9883	0.9698
P_{g8} (MW)	20.7461	21.1087	T_{27-28}	0.9821	0.9777
P_{g11} (MW)	12.1265	12.1602	Q_{c10} (MVar)	4.5972	4.9797
P_{g13} (MW)	12.00	12.0113	Q_{c12} (MVar)	2.5990	2.2098
V_{g1} (pu)	1.0797	1.0825	Q_{c15} (MVar)	5.0	4.9254
V_{g2} (pu)	1.0621	1.0641	Q_{c17} (MVar)	3.8280	4.6838
V_{g5} (pu)	1.0333	1.0332	Q_{c20} (MVar)	4.3910	2.9661
V_{g8} (pu)	1.0362	1.0374	Q_{c21} (MVar)	5.0	4.9994
V_{g11} (pu)	1.0621	1.0819	Q_{c23} (MVar)	2.1225	3.6618
V_{g13} (pu)	1.0544	1.0398	Q_{c27} (MVar)	5.0	4.8890
T_{6-9}	1.0442	1.0314	Q_{c29} (MVar)	3.3295	2.8936

TABLE III
COMPARISON OF PERFORMANCE (IEEE 30 BUS SYSTEM)

Algorithms	Case 1		Case 2		Total FEs
	Cost (\$/h)	Time (s)	Losses (MW)	Time (s)	
GA	800.51	5.5248	3.0943	6.1664	5,050
KAGA	800.48	1.9863	3.0932	1.9630	1,050
PSO	800.43	5.7212	3.0894	5.4236	5,050
KAPSO	800.45	1.8494	3.0948	1.8909	1,050
EEA	800.08	5.9694	3.2823	5.7167	109
BBO	799.11	11.02	-	-	10,000
MSA	800.51	14.91	3.1005	-	5,000
GSO	799.06	12.26	-	-	6,000
PPO	799.06	10.71	2.867	-	9,000

solutions are compared with the results from Efficient Evolutionary Algorithm (EEA) [34], Biogeography Based Optimization (BBO) [35], Moth Swarm Algorithm (MSA) [20], Glow-worm Swarm Optimization (GSO) [36] and Predator–Prey Optimization (PPO) [37] in Table III. The differences between the optimal costs are negligible, but the KAGA and KAPSO clearly outperformed the other algorithms in terms of shorter computing times. Moreover, when MATPOWER power flow calculation was run using the parameter settings given in literature, the voltage magnitudes of load buses of EEA and BBO were above their specified upper limit (1.05 pu), as shown in Fig. 1; this implies that KAGA and KAPSO yield more reliable results as all voltages were within the specified limit. Note also that the upper voltage limit of load buses for GSO and PPO was actually set to 1.1 pu, which is different from other algorithms. In order to make a fair comparison and further verify the proposed algorithms, running KAGA and KAPSO with the limit increased to 1.1 pu produced the optimal costs of 798.97 and 798.92 \$/h,

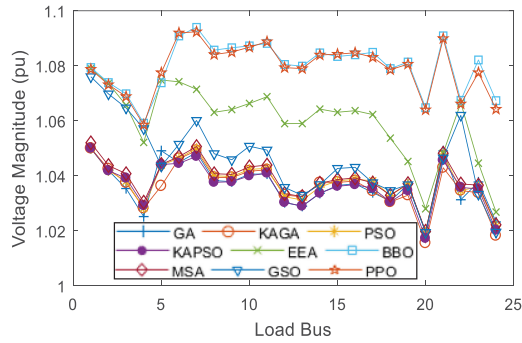


Fig. 1. Voltage profiles of load buses.

TABLE IV
AVERAGE CONVERGENT SOLUTIONS, FUNCTION EVALUATIONS, AND CPU
TIME OF 50 TRIALS FOR MINIMIZING GENERATION COST

Algorithms	Mean	Best	Worst	STD	Time (s)	FES
GA	801.09	800.55	802.37	0.4584	4.8675	3672
KAGA	801.06	800.58	802.68	0.5037	1.9749	794
PSO	800.54	800.44	800.68	0.0627	3.878	3,366
KAPSO	800.65	800.48	800.89	0.0908	1.485	532
EEA	800.17	800.08	800.21	0.0407	5.969	109
BBO	799.19	799.11	799.20	-	8.816	8,000
GSO	799.06	799.05	799.91	-	4.291	2,100
PPO	799.09	799.05	799.13	-	4.280	3,600

respectively, with the KAPSO result being overall the best.

For Case 2 of minimizing active power losses, the proposed Kriging assisted approaches performed as good as GA and PSO, while yielding a better optimum than EEA and MSA. Setting the upper voltage limit to 1.1 pu (i.e., the same as with PPO) for KAGA and KAPSO resulted in the losses of 2.8389 and 2.8333 MW, respectively; again the KAPSO result was overall the best. Even more importantly, however, KAPSO achieves the result with the shortest elapsed computational time and the smallest number of necessary FEs. Note that EEA appears to require fewer FEs, but in reality performs many additional sensitivity matrix calculations at each iteration, thus the computational time is significantly longer than for KAPSO.

In the forgoing analysis the maximum number of iterations was used as a stopping criterion; this may be regarded rather conservative. As an alternative—and to assess the robustness of KAGA and KAPSO—a criterion was applied to terminate iterations when the standard deviation of the fitness of the population/personal best becomes smaller than 0.01; this was tested for the case of minimizing the generation cost. The average, best, worst and standard deviation (STD) values of the optimal solution, as well as the average CPU time of 50 trials and the number of FEs, are compared in Table IV. The GA and PSO required a high number of FEs to converge, but the overall time was still shorter than for other methods reported in literature; this is likely due to the more advanced hardware or more efficient coding, so direct comparisons are somewhat inconclusive. But KAGA and KAPSO have again shown their superior performance being 2.5 and 2.6 times faster than GA and PSO and requiring fewer FEs. In addition, the average CPU times of GA, KAGA, PSO and KAPSO for each iteration were

TABLE V
COMPARISON OF PERFORMANCE (IEEE 118 BUS SYSTEM)

Algorithms	Case 1		Case 2		
	Cost (\$/h)	Time (s)	Losses (MW)	Time (s)	Total FEs
GA	129,647	211.05	10.177	206.23	100,100
KAGA	129,642	84.853	10.305	84.327	20,100
PSO	129,627	175.72	9.929	176.39	100,100
KAPSO	129,619	75.744	9.909	75.960	20,100
MSA	129,640	-	-	-	50,000
GPSO	129,627	328	9.833	-	768,000
TLBO	129,682	1,885	-	-	-

*Note that all algorithms have used the system data from MATPOWER IEEE 118 bus casefile and they obeyed the same constraints specified in the data file.

0.0575s, 0.0039s, 0.0554 and 0.0039s, respectively, thus KAGA and KAPSO were approximately 14 times faster (because only two FEs were required at each iteration). Regarding the solution quality, KAPSO offers a slightly better solution in terms of mean, best, worst and STD values than KAGA, while the consistency of the optimal solution obtained by KAPSO is quite acceptable (small standard deviation). It can therefore be concluded that the proposed KAPSO algorithm provides better and more reliable solutions with faster convergence, less CPU time and fewer FEs. Furthermore, acceleration of the GA and PSO may be accomplished by using Kriging to reduce the total number of FEs.

B. IEEE 118 Bus System

This larger-scale system has 4,242 MW active and 1,439 MVar reactive power demand with 130 control variables, including 53 active power outputs and 54 terminal voltages of generators, 9 transformers and 14 shunt compensators. The upper and lower limits for voltage magnitudes of all buses are set to 1.06 pu and 0.94 pu, while the shunt capacitors can compensate a minimum of 0 to a maximum of 30 MVar. For solving the OPF problems the population size and maximum number of iterations for the proposed algorithms were set to 100 and 10,000, respectively. For this case, the ‘sigma’ described in (31) was set to 0.01 for PSO and KAPSO to achieve better optimal results.

In this particular case, we found that the penalty factor used in IEEE 30 bus system for the load bus voltage and the reactive power generation constraints were no longer adaptive (i.e., the penalties for voltage and reactive power, compared with the objective value, were too small and too large, respectively). As a result, the voltage magnitudes and the reactive power generated were easily pushed beyond the specified limits and trapped at local optima. Therefore, the penalty values of 5,000 and 0.02 for the load bus voltage and the reactive power generation constraints were used instead to deal with the inequality constraint of state variables. Such settings usually lead the algorithms to find good optimal solutions, but the reactive power generation may violate the specified limits because the penalty value is set too small. In order to address such reactive power generation violation, the “Reactive Power Enforcement (RPE)” strategy [33] was employed. The reactive power generation is enforced to its limit and this “PV” bus is re-defined to “PQ” bus for power flow calculations, if any violation is detected at the maximum iteration. This is a very effective strategy to deal with the reactive

TABLE VI
OPTIMAL SOLUTIONS OBTAINED BY KAPSO FOR MINIMIZING GENERATION COST (IEEE 118 BUS SYSTEM)

Bus i	P_{gi} (MW)	V_{gi} (pu)	Bus i	P_{gi} (MW)	V_{gi} (pu)	Bus i	P_{gi} (MW)	V_{gi} (pu)	Bus i	P_{gi} (MW)	V_{gi} (pu)	Line i	T_i	Bus i	Q_{ci} (MVar)
1	24.8704	1.0317	32	15.7959	1.0396	66	349.4961	1.0600	92	5.571E-5	1.0481	8	0.9752	5	3.6844
4	6.359E-4	1.0585	34	2.8739	1.0473	69	454.9231	1.0600	99	1.032E-4	1.0481	32	1.0577	34	9.4424
6	6.882E-4	1.0492	36	8.4881	1.0451	70	0	1.0381	100	231.6288	1.0513	36	0.9857	37	3.372E-5
8	0.0010	1.0380	40	49.3657	1.0355	72	2.421E-4	1.0401	103	38.2816	1.0410	51	0.9873	44	7.8479
10	400.5426	1.0559	42	41.2709	1.0361	73	0.0133	1.0372	104	3.270E-5	1.0309	93	0.9937	45	15.2793
12	85.5304	1.0468	46	19.0478	1.0396	74	17.8543	1.0282	105	5.4196	1.0278	95	1.0035	46	9.9933
15	20.5768	1.0412	49	193.6240	1.0531	76	22.7792	1.0128	107	29.4204	1.0219	102	0.9811	48	7.6838
18	12.2132	1.0428	54	49.4290	1.0303	77	1.848E-5	1.0455	110	6.6117	1.0259	107	0.9533	74	25.5985
19	20.5846	1.0405	55	31.1646	1.0299	80	432.0845	1.0575	111	35.1759	1.0327	127	0.9899	79	29.9952
24	1.365E-5	1.0481	56	32.0223	1.0298	85	5.884E-5	1.0504	112	36.5528	1.0177	-	-	82	29.9261
25	194.3872	1.0600	59	149.5064	1.0494	87	3.5924	1.0533	113	3.568E-4	1.0491	-	-	83	9.3625
26	280.8623	1.0600	61	148.1700	1.0564	89	502.8165	1.0600	116	1.431E-6	1.0600	-	-	105	16.3973
27	10.5999	1.0413	62	1.053E-4	1.0528	90	2.348E-4	1.0417	-	-	-	-	-	107	20.5145
31	7.2851	1.0357	65	352.9077	1.0600	91	3.693E-4	1.0449	-	-	-	-	-	110	7.7709

power generation constraints and usually requires few power flow calculations when compared with the computational effort of searching for a feasible optimal solution within the solution space.

Moreover, the GA, PSO, KAGA and KAPSO were tested for 50 trials to assess their reliability. It was noted that there were some inconsistencies in the answers related to the mean, best, worst and STD values of the 50 trials, a feature not uncommon when dealing with large problems and when using limited computational budget. The KAPSO performed overall best in terms of mean, best, worst and STD values for case 1 and case 2, whereas KAGA obtained larger STD value than GA for case 1 and KAGA performed as well as GA for case 2. In this paper, the best solutions obtained from the 50 trials have been presented. Furthermore, in order to ensure the algorithms would generate truly random values, a MATLAB pseudo-random number generator 'rng('shuffle')' was used at the beginning of the script.

The minimum cost and minimum losses from GA, KAGA, PSO and KAPSO simulations (GA and PSO using population size of 100 and the maximum number of iterations 1,000), compared with MSA, GPSO [38] and TLBO [39] results, are presented in Table V. It can be observed that the Kriging assisted methods perform as good as (or even better than) the GA and the PSO, and again the proposed approaches outperform the GA, PSO and other algorithms in terms of efficiency of computation, both the necessary FEs and actual simulation times. A more detailed inspection of Table V reveals that KAPSO offers the lowest cost, although marginally worse losses than GPSO. However, GPSO required a massive number of FEs and is computationally very inefficient. Similarly, TLBO has been shown to be computationally extremely inefficient (1,885s). Comparing GA and KAGA reveals that the minimum cost and minimum losses are similar, but the KAGA requires less computational effort in terms of CPU time and FEs. Moreover, without going into details, the minimum cost provided by TLBO was achieved while violating some reactive power generation constraints (not negligible) and the MSA's solution could not be verified due to the lack of active power generation value at bus 1. The KAPSO results, on the other hand, are consistently within the imposed limits (see Fig. 2 and Fig. 3) thus the algorithm appears to be very robust.

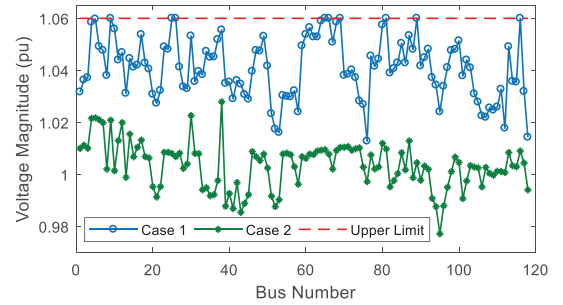


Fig. 2. Voltage profiles of all buses for KAPSO.

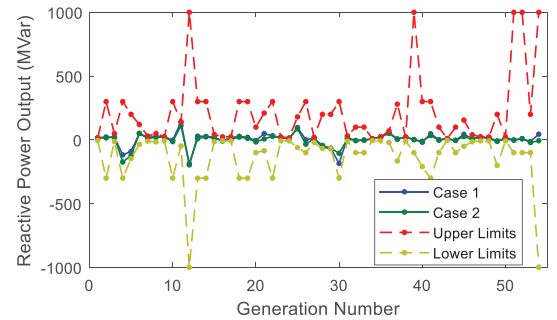


Fig. 3. Reactive power generations for KAPSO.

Finally, Table VI shows a complete set of results for the optimal solution. When the MATPOWER power flow simulation was run using the approximate data given in the table, the reactive outputs at buses 1, 76 and 85 were 15.1191, 23.0305 and 23.0718 MVar, respectively, which violated the upper limits of 15, 23 and 23 MVar. However, such violation does not happen when the exact solutions obtained by KAPSO are used. Table VI by itself does not show anything particularly interesting, but is included for the sake of completeness to allow reproduction of the results.

V. CONCLUSION

In this paper, a Kriging assisted strategy has been proposed to expedite the evolutionary algorithms for the solution of optimal

power flow (OPF) problems. The inherent difficulty of Kriging formulations due to the requirement of matrix inversion—particularly troublesome for large problems—has been alleviated by applying constant sample size. To balance exploitation and exploration when selecting a particle with a maximum expected improvement value for Kriging, an efficient Euclidean distance based method has been proposed. Before applying to power system optimization, two approaches combining Kriging with GA and PSO, called KAGA and KAPSO, respectively, were tested using four complex unconstrained benchmark optimization functions. The results demonstrated that by providing this assistance by Kriging the total number of necessary function calls has been reduced, sometimes significantly, while the solution quality has been improved.

Consequently, KAGA and KAPSO were applied to the IEEE 30 and 118 bus systems to solve two basic OPF problems of minimizing the generation cost and active power losses. The results shown that the solution quality of the KAGA is as good as the GA, but the computational efficiency of GA in solving OPF problems has been improved by Kriging. For both test cases KAPSO demonstrated consistent and robust performance avoiding violation of constraints (associated with other methodologies) and offering good quality solutions with significant improvements in terms of computational efficiency. It is therefore concluded that the proposed approach is very promising for improving the computation efficiency of evolutionary based algorithms when solving OPF problems. Finally, it could be further studied by combining Kriging with other evolutionary based methods and applying the algorithms to other complex OPF problems with larger system sizes.

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